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ORIGINAL ARTICLE

Performance Analysis of Hybrid Energy Storage Systems in Abandoned Mines: Peak-Shaving and Load Balancing using Multi-Source Heat Storage

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Abstract
This paper presents a multi-source thermal storage for peak shaving and load balancing to improve the performance of Hybrid Energy Storage (HES) systems for abandoned mines. The proposed system combines renewable energy sources, such as solar, wind, and traditional power sources, and use Information Gap Decision Theory (IGDT) to optimize scheduling under uncertainties. Furthermore, it implements a Multi-Source Thermal Storage Peak Shaving (MSTSPS) with a Load Balancing Strategy (LBS) to manage variable loads efficiently, which significantly reduces peak-to-valley load differences and scheduling costs. In the end, it presents a case study using IEEE-39 nodes to evaluate the performance of the proposed work. Results show that the proposed system reduced peak-to-valley load differences with substantial cost savings, highlighting the potential of abandoned mines as sustainable, cost-effective storage solutions for renewable energy.

Keywords Abandoned mines · Hybrid energy storage · Thermal storage · Peak shaving · Scheduling cost

1 Introduction

Recent years have observed a huge spike in energy demand worldwide, which motivates researchers to create environmentally friendly and sustainable energy solutions [5, 17]. Therefore, the world is gradually moving towards Renewable Energy (RE) sources like solar, hydro, or wind [4, 7]. However, these researchers often overlook the potential of abandoned mines as a RE source.

With the rapid development of the social economy, the mining industry plays a pivotal role in fields such as energy, transportation, and urban construction [26]. However, the waste generated by mining causes severe damage to the environment [21]. Furthermore, once mining resources are depleted, vast underground spaces with degraded ecosystems and surface areas are left, known as Abandoned Mines [25]. The World Business Council for Sustainable Development (WBCSD) states that millions of abandoned mines exist worldwide. Therefore, managing abandoned mines is essential for environmental protection and restoration [13]. In contrast to environmental challenges, these mines also present a unique and mainly untapped opportunity because they contain a large amount of RE sources [37]. By utilizing these energy sources, dependence on traditional energy can be effectively reduced, energy costs can be lowered, and energy utilization efficiency can be improved [33].

The underground cavities and open surfaces of abandoned mines make them great energy generation and storage sources. Winde et al. [35] show that such sites can host various renewable energy systems. For example, in addition to RE and Energy Storage Systems (ESS), abandoned mines such mines can be utilized to construct geothermal Power Generation Systems (PGS), Solar Thermal Power

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Generation (STPG) systems, and small hydropower stations [14, 23, 35]. Furthermore, the natural characteristics of mines, such as stable underground temperatures and isolated locations inside the mine, can help optimize the performance and lifespan of energy storage technologies. All these benefits of abandoned mines allow the integration of Hybrid Energy Storage (HES) systems capable of managing multiple renewable sources [12].

The main objective of the HES system is to combine and store energy from multiple sources [16]. Although the HES system for abandoned mines has achieved certain results in practical applications, many problems still urgently need to be solved [15]. Firstly, the coupling of multiple energy sources is a distinguishing feature of hybrid energy systems from traditional systems, posing new challenges to optimizing system operation. For example, the load and output demands of different energy sources are often highly variable and uncertain, which traditional systems often fail to handle. To handle this problem, we present a Multi-Source Thermal Storage Peak Shaving (MSTSPS) with a Load Balancing Strategy (LBS). The MSTSPS uses a scheduling algorithm to address these challenges by integrating different energy sources. Furthermore, we use information gap decision theory (IGDT) to overcome variabilities caused by different sources and enhance the reliability of the proposed system.

Secondly, integrating HES systems in abandoned mines for practical applications also needs attention [1]. This study's innovation lies in considering the uncertainty of Wind and Solar (W&S) loads and using IGDT to optimize scheduling, achieving the target of minimizing the system's Total Scheduling Cost (TSC), making it suitable for practical applications. Furthermore, by introducing the LBS, the problem of excessive load on the distribution network caused by the integration of RE in the HES system of abandoned mines has been solved. The key contributions of this study are listed as follows:

- We present the HES system using IGDT to optimize the scheduling of solar, wind, and traditional power sources in abandoned mines. Furthermore, IGDT also handles uncertainties in (W&S) output, making the proposed framework more reliable.
- We use MSTSPS with LBS to reduce peak-to-valley load differences in abandoned mines. Our findings show that the proposed work reduces peak-to-valley load differences from 120 MW to 90 MW.
- We evaluated various optimization methods to reduce scheduling costs over conventional methods, even under volatile load conditions.
- We performed a case study using IEEE-39 nodes to validate the effectiveness of the scheduling strategy.

The rest of the paper is organized as follows: Section 2 provides an overview of related studies and the novelty of the proposed work. Section 3 presents the proposed HES system for abandoned mines, while Sect. 4 evaluates the proposed work's performance with a case study. Finally, Sect. 5 concludes the work.

2 Literature Review on Existing HES Systems for Renewable Energy Management

The energy system converts energy from the source into usable forms through various energy conversion and transmission technologies and delivers them to end users, such as industrial, residential, and commercial sectors [18]. With the increase of diverse RE sources, such as hydro, wind, and solar, building energy systems capable of handling/storing energy from all these sources is important. Therefore, the demand for HES systems has grown in recent years due to their ability to manage data from multiple sources and handle their variability [16]. Researchers have explored different techniques to enhance the performance of HES systems depending on the requirements, such as energy conversion efficiency, improving storage capabilities, and making them cost-effective.

Abdelghany et al. [3] proposed an economic/environmental microgrid control strategy considering the cost of hydrogen equipment and battery degradation. This strategy used scenarios to reflect the randomness of RE and load demand and managed electricity market participation through interaction with the primary grid and external consumers. Their proposed controller is effective in the presence of uncertainty and could effectively manage integrated systems. Abdalla et al. [2] used prediction and adaptive smoothing mechanisms to avoid arbitrary selection of filtering time constants, thereby reducing the pressure on ESSs. In performance testing, the proposed controller performed better than smoothing techniques and slope control based on fixed time constants under prediction errors [2].

Samadhiya et al. [30] proposed an energy management and control scheme for a hybrid AC-DC microgrid based on solar photovoltaics as the main power source. This scheme achieved efficient power management and DC voltage regulation by using dual ESSs and three different types of converters. Their proposed scheme could effectively control the voltage ripple of the DC link, reduce harmonic distortion in the grid current, and improve power quality. In contrast, Khazali et al. [20] proposed a microgrid capacity planning framework built on RE and hybrid battery ESSs. This framework adopted a Mixed Integer Linear Programming (MILP) method, taking into account the constraints of battery aging and annual unmet load. The cost of their

proposed HES was lower than that of single battery technology. Furthermore, the authors demonstrated that the use of hybrid systems could effectively reduce the capacity degradation of lead-acid batteries, reducing the risk of system replacement and degradation.

Energy system scheduling refers to the management and optimization of energy production, transmission, and distribution to guarantee the safety, stability, and sustainability of energy supply while improving the efficiency and economy of the energy system [8]. Meiers et al. [24] used two sets of electric vehicles and two fixed battery storage systems on the campus of Saarbrücken Saar University in Germany as examples to explore the potential of using photovoltaic power stations and intelligent charging technology for peak shaving. Compared with reference systems without battery storage capacity and photovoltaic power stations, combining electric vehicle fleets with fixed battery storage systems could effectively improve peak shaving potential and reduce total electricity costs. Zhang et al. [39] proposed a model for evaluating the peak shaving capability of central air conditioning loads in RE-dominated power systems and validated it on multiple time scales. This model could quickly respond to peak-shaving commands in the power system, optimize system resource allocation, and improve peak-shaving capabilities. It has been evaluated and optimized in different scenarios, confirming its effectiveness [39].

Qiu et al. [27] developed a coordinated Scheduling Model (SM) for electric hydrogen HES integrated energy systems grounded on two-stage distributed robust optimization. The model was divided into two stages: pre-day optimal scheduling under deterministic prediction information and intra-day rescheduling under a distributionally robust optimization method, considering both economic operation and carbon emissions. Compared to traditional methods, this model could better balance conservatism and robustness, achieving optimal scheduling tasks [27]. Rana et al. [28] confirmed the effectiveness of the classification variable decision tree algorithm by comparing its peak shaving performance with the general capacity addition technique in an independent microgrid system. This algorithm could better minimize peak demand when dealing with hybrid photovoltaic cell ESSs, thereby achieving energy conservation, reducing system losses, and lowering operation and maintenance costs.

Previous studies provide valuable insights into the architecture, scheduling, and variation management in HES systems. However, several key problems exist that require attention. For example, most studies do not consider optimization and evaluation in different scenarios for peak shaving in the power system. Furthermore, these studies have not explored the viability of HES systems in complex environments, such as abandoned mines. Another significant drawback of most studies is using LSB to address the fluctuations in HES systems. To address these challenges, this study proposes a HES system for abandoned mines using MST-SPS and LBS combined with IGDT for effective uncertainty management. The proposed system aims to enhance energy usage efficiency, reduce energy costs, and promote the effective utilization of abandoned mining resources. A comparison of the proposed method with existing studies is illustrated in Table 1.

3 Proposed HES System of Abandoned Mines

3.1 HES System Architecture

There are a large number of mining-induced fractures and goaf spaces underground in abandoned mines. These spaces can provide sufficient storage space for the HES system. In the HES system of abandoned mines, the thermal storage photovoltaic power station has the ability to convert solar energy. By optimizing the photovoltaic thermoelectric conversion process through a Thermal Storage System (TSS), its output power is flexible and schedulable [10]. Based on this, the proposed HES system integrates solar, wind, and hydro resources with TSS that helps convert light, heat, and electrical energy into an electricity supply, as shown in Fig. 1

The basic structure of the HES system constructed in this study mainly consists of three core parts: the concentrated

Table 1 Comparison of the proposed and existing HES systems for RE management

Method	Peak Shaving Capability	Power Management	Optimal Scheduling	Cost Effectiveness	Applicability in Mines
[3]	✓	✓	X	✓	X
[2]	X	✓	X	X	X
[30]	X	✓	X	✓	X
[20]	X	X	✓	✓	X
[24]	✓	✓	X	X	X
[39]	✓	✓	✓	✓	X
[27]	✓	✓	✓	X	X
[28]	✓	X	X	X	X
Proposed Work	✓	✓	✓	✓	✓

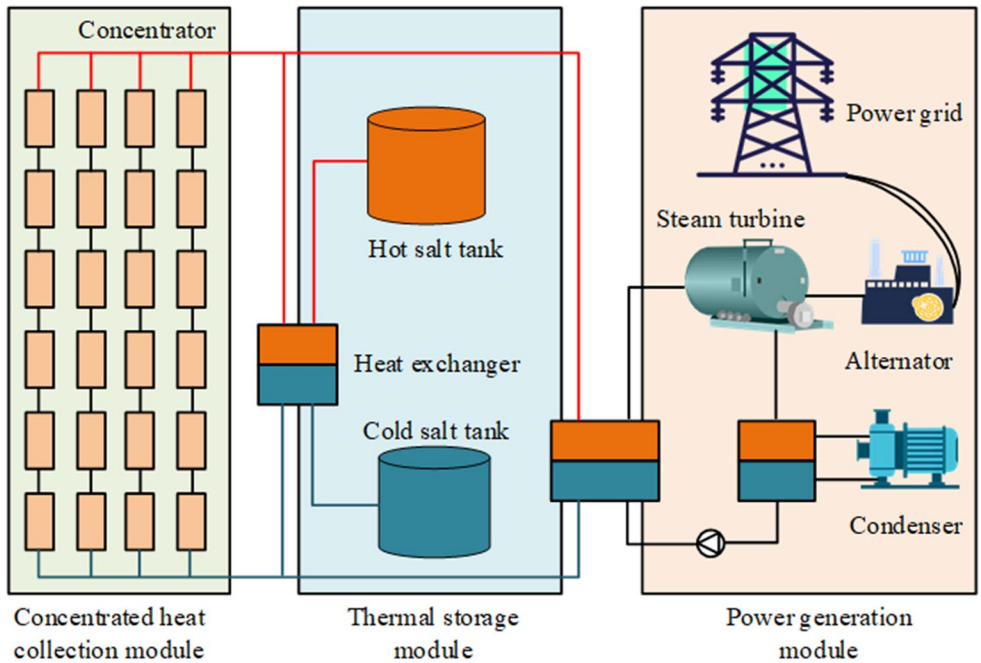

Fig. 1 Basic structure of the proposed HES system

heat collection module (M1), the TSS (M2), and the Power Generation System (PGS, M3), as shown in Fig. 1. In the proposed HES system, these three modules work together to convert solar energy into electricity to meet the high need for sustainable energy. The process starts at M1, whose primary function is to collect solar energy through spotlight technology and convert it into thermal energy. The energy is later stored in the TSS of M2, which is responsible for storing the collected thermal energy for energy output at the appropriate time [36]. Finally, the M3 converts stored thermal energy into electrical energy and supplies it to the power grid. The heat transfer fluid plays a crucial role in efficiently transferring and converting energy between these three parts [6]. In addition, the system has peak shaving capability, which can effectively balance the load of the power grid.

The main goal of this work is to improve energy efficiency by capturing solar energy and converting it into thermal energy, which is then converted into electrical energy using a generator set. Inside the power station, the three types of energy flow - light, heat, and electricity - are converted into each other to achieve maximum energy utilization. The HES system utilizes batteries, supercapacitors, and other devices to store and convert various forms of energy [32]. When there is sufficient sunlight, solar panels convert light energy into electrical energy, with some directly used for power supply and others stored for future use. In contrast, if there is insufficient light, the stored electrical and thermal energy can be converted into each other to satisfy the electricity requirement. In the thermal energy conversion process, solar water heaters or STPG systems convert light energy into thermal energy. This part of the heat energy can be utilized for Power Generation (PG) or stored for other purposes. In converting electrical energy, the generator set converts thermal energy into electrical energy to meet the electricity demand.

The system operates in six modes depending on load requirements and resource availability. For example, M2 releases heat to generate electricity during insufficient light or nighttime periods. In contrast, when the energy demand is low, all three subsystems enter an idle state to conserve energy. This energy flow relationship ensures that the HES system can operate efficiently under both sufficient and insufficient lighting conditions, ensuring its reliability. The system's power balance relationship is shown in Eq. (1).

$$\begin{cases} P_t^{SF} = \eta_{s-th} P_t^S \\ P_t^{SF} - P_t^{PB} - P_t^{cha} + P_t^{dis} - P_t^{spill} = 0 \\ P_t^{CSP} = \eta_{ht-p} P_t^{PB} \end{cases} \tag{1}$$

In Eq. (1), during period t, the solar radiation conversion heat power is P_t^{SF}, the photothermal conversion efficiency of the device is η_{s-th}, and the illumination power is P_t^S. The thermal power transported by the heat transfer fluid is P_t^{PB}. The heat storage power is P_t^{cha}, and the heat release power is P_t^{dis}. The discarded thermal power is P_t^{spill}. The output power of STPG is P_t^{CSP}. The energy equation of the TSS is shown in Eq. (2).

$$E_{mi}^{HSS} \leq E_{i}^{HSS} \leq E_{ma}^{HSS} \tag{2}$$

In Eq. (2), during period t, the thermal energy stored in the TSS is E_i^{HSS}. Its minimum and maximum heat storage capacities are $E_{\mathrm{mi}}^{\mathrm{HSS}}$ and $E_{\mathrm{ma}}^{\mathrm{HSS}}$, respectively. In the PGS, the thermal PG cycle is similar to conventional generation, both of which generate electricity by driving a steam turbine with the constraint conditions shown in Eq. (3).

$$x_{j,t}P_j^{\mathrm{CSP\text{-}mi}} \leq P_{j,t}^{\mathrm{CSP}} \leq x_{j,t}P_j^{\mathrm{CSP\text{-}ma}} \tag{3}$$

In Eq. (3), the start-stop state of the solar thermal generator unit j during period t is $x_{j,t}$. Its output value is $P_{j,t}^{\mathrm{CSP}}$, with a maximum output value of $P_j^{\mathrm{CSP\text{-}ma}}$ and a minimum output value of $P_j^{\mathrm{CSP\text{-}mi}}$. The equivalent load calculation of conventional Thermal Power Units (TPU) is shown in Eq. (4).

$$P_{\mathrm{mi}}^{G} \leq L_{\mathrm{eq}} = L - P^{R} \leq P_{\mathrm{ma}}^{G} \tag{4}$$

In Eq. (4), L_{eq} is used for the equivalent load, and L represents the original load. P^R denotes clean energy output. The lower limit of output for TPUs is P_{mi}^{G}, and the upper limit of output is P_{ma}^{G}. The internal energy flow relationship of the HES system is shown in Fig. 2.

Load Side Demand Response (LSDR), as a key adjustable resource, plays a vital position in ensuring the safe, stable, and economically efficient operation of the power system [19]. In scenarios where Wind Power (WP) fluctuates greatly, the demand response mechanism is combined with highly dispatchable Solar Thermal Power Plants (STPP) to achieve stable output and complementarity of WP. At the same time, by guiding users to adjust their electricity consumption plans, increasing the proportion of RE in the power system, and reducing dependence on traditional energy, the economic operation level of the system can be improved. By utilizing demand response mechanisms and the high dispatchability of STPPs, stable output and complementarity of WP can be achieved, thereby reducing the dependence of the electricity market on traditional energy sources and improving operational efficiency [31]. After adding LSDR, the equivalent load of the TPU is shown in Eq. (5).

$$L'_{eq} = L - P^{R} + D \tag{5}$$

In Eq. (5), the equivalent load at this time is L'_{eq}, and the LSDR quantity is D.

3.2 MSTSPS and LBS

The HES system for abandoned mines is a system that comprehensively utilizes multiple energy storage technologies, complementing the benefits of various energy storage approaches to complete efficient energy storage and management. The system adopts multi-source thermal storage technology, which can balance energy demand and supply under different load conditions, thereby improving overall energy storage efficiency. In the Multi-Source Joint Scheduling (MSJS) model, the abandoned mine HES system is divided into several parts: WP, Solar Thermal Power Station (STPS), ESS, and hydropower station. On the one hand, thermal storage STPSs can be adjusted according to WP and light intensity to utilize clean energy efficiently. On the other hand, ESSs and hydropower stations can be scheduled according to load demand to ensure safe and stable operation [22]. The objective function of scheduling cost in the system is shown in Formula (6).

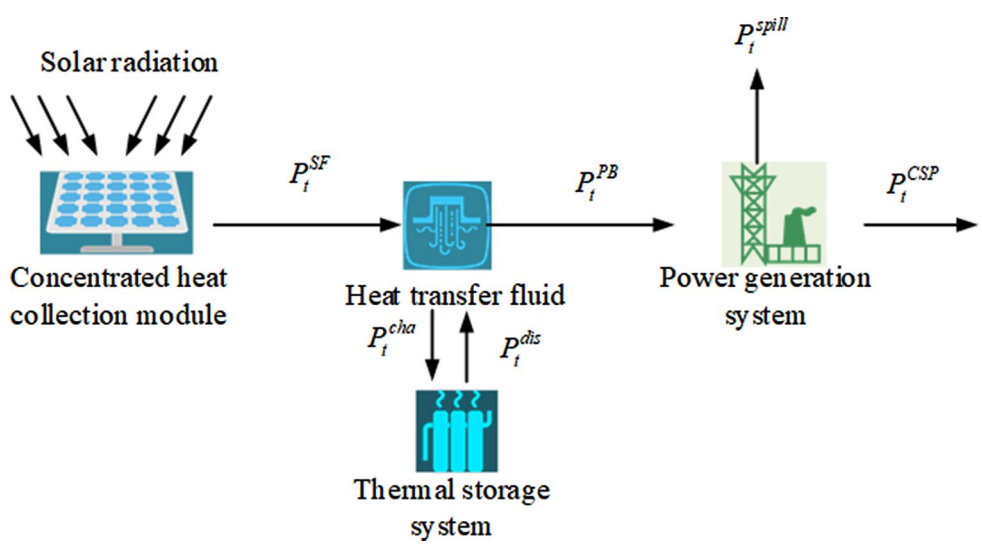

Fig. 2 Internal energy flow relationship of the HES system

$$F = \min(F_g + F_{ss}) \tag{6}$$

In Eq. (6), the TSC of the system is F, the fuel cost of the conventional unit is F_g, and the total cost of starting and stopping the unit is F_{ss}. The constraint conditions for system power balance are shown in Eq. (7).

$$\sum_{i=1}^{N_G} P_{i,t}^{G} + \sum_{j=1}^{N_{CSP}} P_{j,t}^{CSP} + P_t^{W} = L_t \tag{7}$$

In Eq. (7), during period t, the output of conventional unit i and solar thermal generator unit j are $P_{i,t}^{G}$ and $P_{j,t}^{CSP}$, and the gross of scheduling periods is N_G and N_{CSP}. During the t period, the WP processing is P_t^{W} and the load demand is L_t. The constraint conditions for conventional unit processing are shown in Eq. (8).

$$u_{i,t} P_i^{G-mi} \le P_{i,t}^{G} \le u_{i,t} P_i^{G-ma} \tag{8}$$

In Eq. (8), the start stop state of conventional generator set i during period t is $u_{i,t}$, with an output value of $P_{i,t}^{G}$, a maximum output value of P_i^{G-ma}, and a minimum output value of P_i^{G-mi}. The rotation backup constraint is Eq. (9).

$$\begin{aligned} &\sum_{j=1}^{N_G} \left(u_{i,t} P_{i,t}^{G-ma} - P_{i,t}^{G}\right) \\ &\quad + \sum_{j=1}^{N_{CSP}} \left(x_{j,t} P_{j,t}^{CSP-ma} - P_{j,t}^{CSP}\right) \ge \rho \left(P_t^{W} + L_t\right) \end{aligned} \tag{9}$$

As explained earlier, the MSTSPS aims to minimize TSC by coordinating the dispatch of conventional, solar thermal, and wind sources and thermal energy storage. In addition, LBS helps smooth out the load profile using a peak shaving and valley filling mechanism through controlled charging and discharging of ESDs. The peak shaving goal can be described using a modified load equation:

$$L_t' = L_t - OUT_{t,p}(t) + IN_{t,p}(t) \tag{10}$$

where L_t' is the adjusted net load at time t, and $OUT_{t,p}(t)$ and $IN_{t,p}(t)$ are the discharge and charge rates of the ESD, respectively. The system aims to minimize the PVD in the adjusted load using the following objective function:

$$\min \left(\max_t L_t' - \min_t L_t'\right) \tag{11}$$

This approach ensures that excess energy during off-peak hours is stored and released during high demand, reducing pressure on the conventional grid and enhancing the grid's stability.

Compared with traditional PG units, wind, and photovoltaic power output is affected by various factors, including randomness, volatility, and intermittency. In MSJS, the first step is to set weight coefficients for the uncertainty of W&S loads to reflect the level of attention of decision-makers to various uncertain factors. Then, using an envelope constraint model, an uncertainty model is established to describe the information gap between predicted and actual values. On this basis, two performance equations, risk avoidance, and risk preference, are constructed to meet the different needs of different decision-makers when facing uncertainty.

After solving the MSJS model, all parties involved need to carry out actual execution according to the scheduling plan issued by the scheduling center. Conventional units, also known as traditional PG equipment, adjust their operating status and output power according to instructions and plans issued by the dispatch center. Wind farms and STPPs must take corresponding adjustment measures based on the actual and planned output [40].

The MSJS process is shown in Fig. 3, which aims to minimize the system TSC. The scheduling process begins with Step 1, which obtains the initial scheduling plan and uses its optimal objective function value as the benchmark. In Step 2, the model incorporates weight coefficients to account for the uncertainty in wind and solar (W&S) loads. The proposed system aims to minimize or maximize overall uncertainty based on the strategy, such as risk-averse (robust scheduling) or risk-seeking (opportunity scheduling). After obtaining the model results, each participant executes the scheduling plan issued by the scheduling center.

Various power plants operate according to scheduling plans when implementing risk mitigation strategies. On the other hand, opportunity scheduling is more aggressive and relies mainly on wind farms and STPPs. However, it typically requires minimal adjustments to actual output. At this point, the conventional unit output plan provided by the scheduling scheme can serve as a reference for actual operation.

The LBS complements this approach by integrating RE sources with Energy Storage Devices (ESD) such as batteries to achieve a stable power supply. This study proposes a scheduling algorithm based on load balancing to address the problem of excessive load on the distribution network caused by the integration of RE in the HES of abandoned mines. The main goal of the proposed algorithm is to keep the distribution network in a high-load operating state while dealing with a large amount of accessed data. The specific approach uses methods such as peak shaving and valley

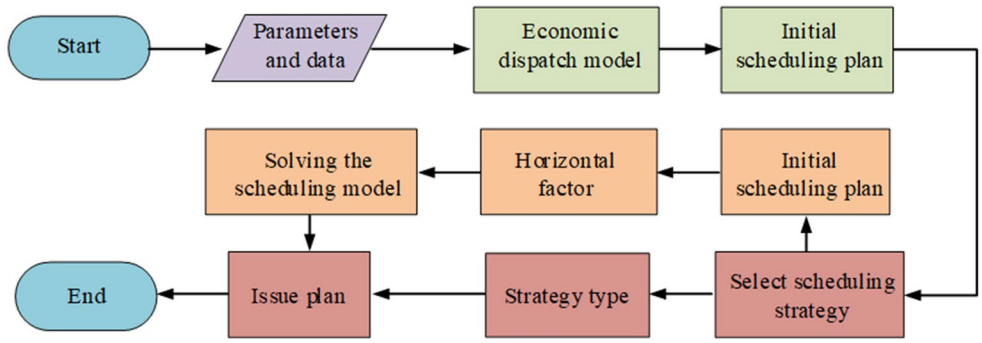

Fig. 3 Multi-source joint scheduling process

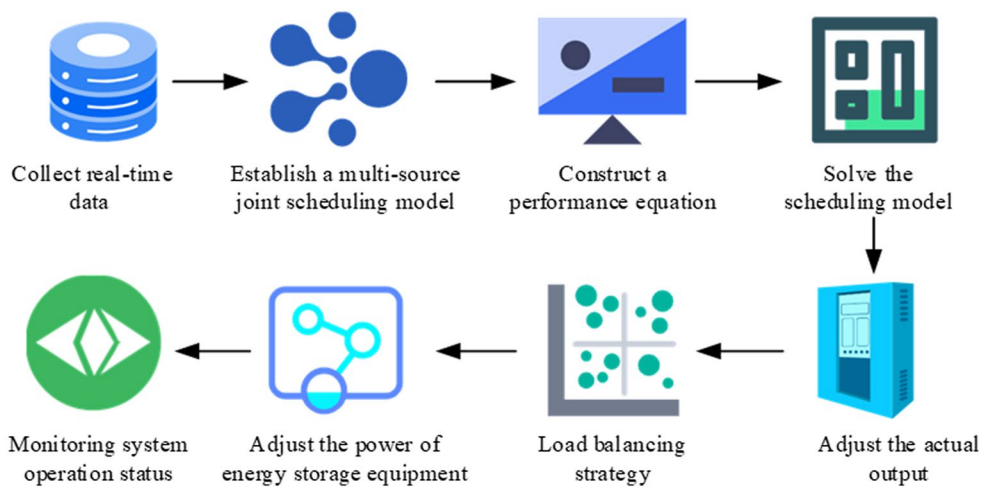

Fig. 4 The complete steps of MSTSPS and LBS

filling to store the electrical energy generated during the peak hours of sunlight through batteries and then release it at night to reduce the peak load level.

At this point, user demand constraints are shown in Eq. (12).

$$H_{mn} \times d_{mn} \leq k_{mn} \tag{12}$$

In Eq. (12), the user demand constraint matrix is H_{mn}, the user demand set is d_{mn}, and the cost matrix of the PG module is k_{mn}. To ensure the maximum PG potential of solar panels, it is necessary to set the highest and lowest charging and discharging values for ESDs under the performance and discharge constraints, as shown in Eq. (13).

$$\begin{cases} 0 \leq IN_{t,p}(t) \leq IN_{t,ma} \\ 0 \leq OUT_{t,p}(t) \leq OUT_{t,ma} \end{cases} \tag{13}$$

In Eq. (13), the maximum charging value is $IN_{t,ma}$, and the maximum discharging value is $OUT_{t,ma}$.

Based on the above content, the complete steps of MSTSPS and LBS are obtained, as shown in Fig. 4. The process starts with collecting real-time data fed to the MSJS model. The system creates a performance equation based on the data to evaluate and optimize the model. After that, the scheduling model is solved, and the output is adjusted to find the best operational strategy. The next step implements load balancing to manage fluctuations, and the power of energy storage equipment is adjusted as required. Finally, the system's operation status is monitored to ensure stability and efficiency. This method ensures that the proposed system optimizes energy usage and reduces costs by integrating multiple renewable sources and storage technologies, enhancing grid stability through effective load balancing. Although the MSTSPS and LBS solutions provide effective models for coordinated load balancing and energy dispatch, both schemes can largely be influenced by the uncertainties from renewable energy resources. To that effect, in the next section, the IGDT is embedded in the scheduling model to enhance robustness from uncertain solar and wind fluctuations.

3.3 Integration with IGDT Under Uncertainty

The proposed scheduling framework integrates the IGDT to address uncertainties associated with wind and solar energy outputs, which is widley used in literature [11, 34]. The primary sources of uncertainty in this system are wind speed and solar irradiance, which directly affect the power outputs of wind farms and solar thermal power stations, respectively

[29]. The main objective of IGDT is to maximize the robustness of the scheduling plan under incomplete or imprecise data.

Let $\bar{P}_t^W$ and $\bar{P}_t^{CSP}$ represent the predicted outputs of wind and solar thermal sources, respectively. The actual values $\tilde{P}_t^W$ and $\tilde{P}_t^{CSP}$ are assumed to lie within an uncertainty set determined by a horizon α:

$$\begin{aligned} &\tilde{P}_t^W \in \left[\bar{P}_t^W(1-\alpha), \bar{P}_t^W(1+\alpha)\right], \\ &\tilde{P}_t^{CSP} \in \left[\bar{P}_t^{CSP}(1-\alpha), \bar{P}_t^{CSP}(1+\alpha)\right] \end{aligned} \tag{14}$$

The IGDT objective is to find the maximum uncertainty horizon α such that the total scheduling cost F remains below an acceptable threshold F_{acc}:

$$\max_{\alpha} \quad \alpha \quad \text{subject to} \quad \max_{\tilde{P}_t^W, \tilde{P}_t^{CSP}} F(\tilde{P}_t^W, \tilde{P}_t^{CSP}) \le F_{\text{acc}} \tag{15}$$

This formulation provides a risk-averse schedule plan. In contrast, a risk-seeking opportunity scheduling tries to minimizes cost at the expense of tolerating a pre-specified level of uncertainty. The MSTSPS schedule plan responds to the computed α by adjusting renewable dispatch and storage utilization in proportion. LBS responds by charging or discharging energy storage appropriately based on the revised net load given the uncertain renewable output. Cumulatively, these steps ensure robustness, reliability, and improved operational economics of the HES system in mines. The performance of the proposed system is evaluated in the next section.

4 Performance Analysis of HES System in Abandoned Mines

The main objective of this study is to evaluate the performance of the proposed HES system in abandoned mines. In this study, the HES system combines different RE sources, handles variations due to multiple sources, and performs optimization/scheduling to make the power grid more stable and efficient. This section evaluates how well the proposed system can reduce peak energy usage, balance the energy load, and reduce overall scheduling costs. However, it is important to understand the characteristics of HES systems and the need for scheduling strategies in such systems.

4.1 Characteristics Analysis of HES System

The HES systems deployed in abandoned mines use these mines as underground storage spaces to integrate different RE sources. Fig. 5 illustrates a Typical Dispatch Daily Load Curve (TDDLC) and highlights the impact of the WP grid connection. Fig. 5(a) plots the typical daily original load curve and equivalent load. The peak value of the original load curve is 430 MW, whereas the valley value is 310 MW, resulting in a Peak Valley Difference (PVD) of 120 MW. The peak value of equivalent load a is 320 MW, the valley value is 140 MW, and the PVD is 180 MW. After all WP is connected to the grid, the system's peak shaving pressure increases. Fig. 5(b) plots the results after WP integration, where load fluctuations become partially smoothed with a max value of 190 MW and a min value of 80 MW. However, relying solely on a WP grid connection for joint scheduling has limited optimization effects. Furthermore, WP alone is insufficient to meet peak shaving needs due to the WPs variability. These limitations motivate using the combination scheduling of multiple energy sources to fully utilize various energy resources and achieve a stable load profile.

Fig. 6 shows the equivalent load curve of TDDLC after adding a solar thermal grid connection that provides stable power during daylight hours, resulting in a smoother curve. Fig. 6(a) demonstrates that after the solar thermal dispatch, the PVD of the equivalent load c is 90 MW, which reduces the load's PVD and alleviates the pressure of peak shaving. Fig. 6(b) shows that the solar thermal output curve is relatively stable, mainly concentrated during the daytime. The primary reason is that solar energy directly shines on the solar collector, which can generate enough heat to drive the generator set during the daytime. In contrast, WP output is affected by wind speed and direction, so its output curve

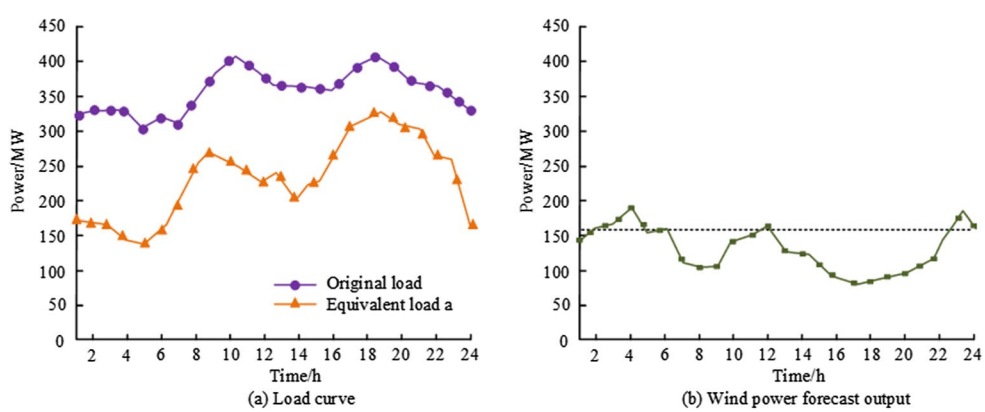

Fig. 5 TDDLC and WP grid connection impact results

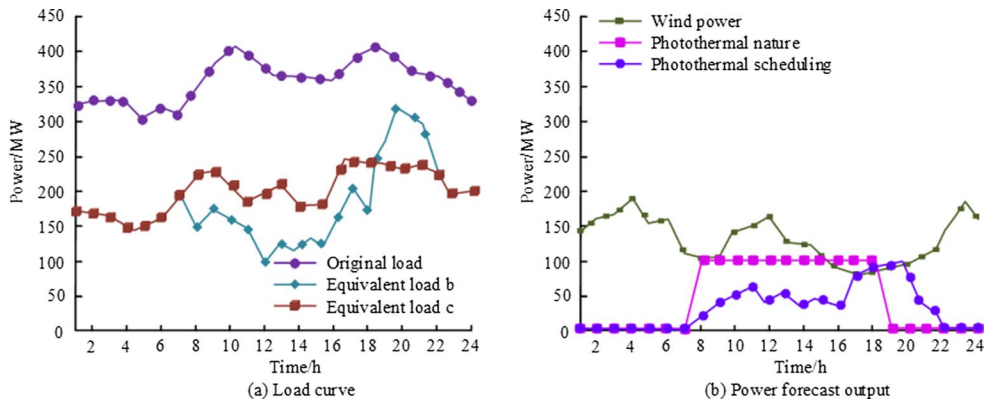

Fig. 6 TDDLC and WP solar thermal grid equivalent load curve

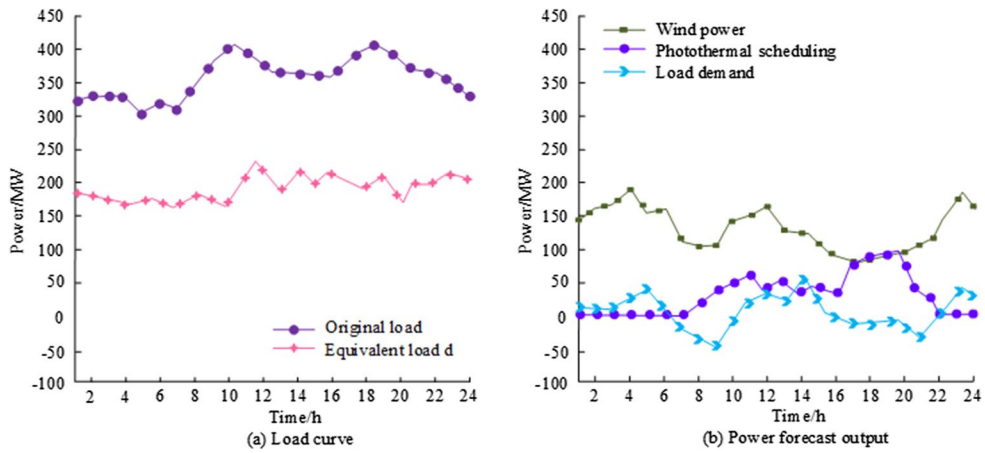

Fig. 7 TDDLC and WP solar thermal grid equivalent load curve

is relatively volatile and may undergo significant changes in a short period.

Fig. 7 emphasizes the advantages of LSDR, which helps in aligning power supply with fluctuating demand. After LSDR, the equivalent load d becomes smoother with a PVD of 55 MW, as shown in Fig. 7(a). Similarly, Fig. 7(b) illustrates the output and scheduling curves, where LSDR adjusts WPs output and scheduling curves. Results show that when WP and solar thermal output increase, LSDR also increases accordingly to meet higher electricity demand. On the contrary, when W&S power output decreases, LSDR will also decrease to avoid excess power supply. In this way, RE such as WP and solar thermal energy can be better integrated into the power system, improving the utilization rate of RE and reducing dependence on traditional fossil fuels.

4.2 Case Study on Scheduling Strategy Evaluation

To validate the effectiveness of the scheduling strategy, we conducted a case study on the HES system of an abandoned mine in Xinjiang, China. The experiment used IEEE-39 nodes, including a 400 MW wind farm, a 100 MW STPS, and 10 Thermal Power Units (TPUs) [38]. The major reason for selecting IEEE-39 nodes is that they have widespread application in power systems literature to evaluate scheduling, stability, and multi-source dispatch policies [9, 38]. It provides a balanced level of complexity and computational tractability, thus being suitable for the suggested hybrid scheduling scheme. In implementation of the model, certain simplifying assumptions were used: load was uniformly distributed on nodes; weather inputs to impact renewable generation were assumed constant for specified areas; and user-side behavior was assumed deterministic without dynamic feedback. These assumptions were necessary in order to keep the impact of the MSTSPS and IGDT-based strategies isolated while keeping the situation comparable and reproducible.

The initial capacity of the TSS is 250 MW. The load demand forecasting curve is shown in Fig. 8. There are two peak load demands in the morning and evening, during which the W&S powers are relatively low. During these two periods, the output of wind and photovoltaic PG cannot meet the load demand and requires to be filled up by other energy sources.

This study uses the MSJS model to solve the TSC that achieves an optimal value of 542,135 ¥. Fig. 9 is the output and thermal curve of the unit. Fig. 9(a) shows the output of the system's various units. The unit's output is the total output of conventional TPUs and solar thermal generators, which can track the load demand curve well. During peak

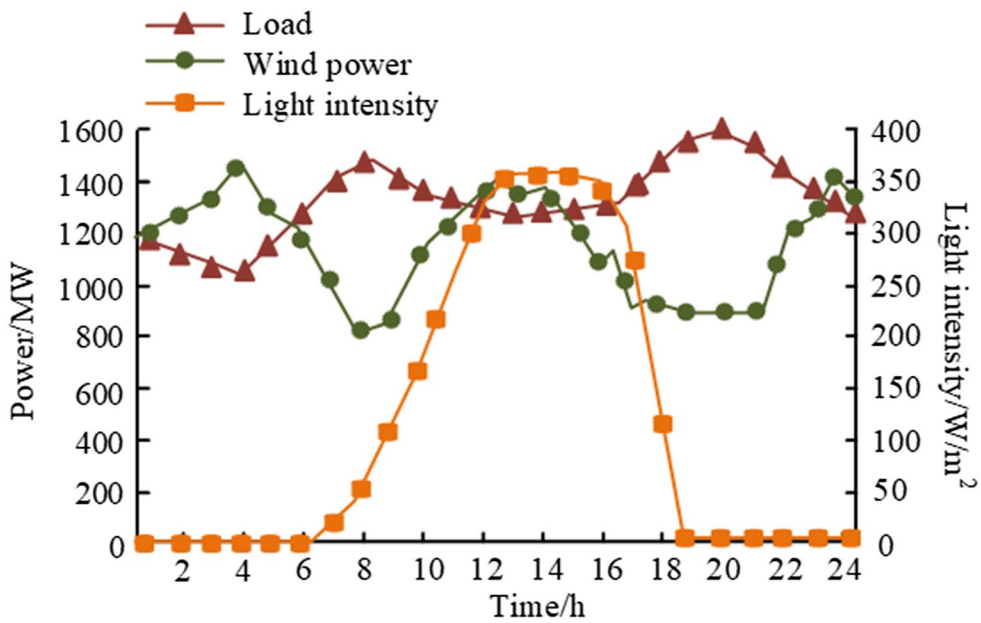

Fig. 8 Load demand forecasting curve of this system

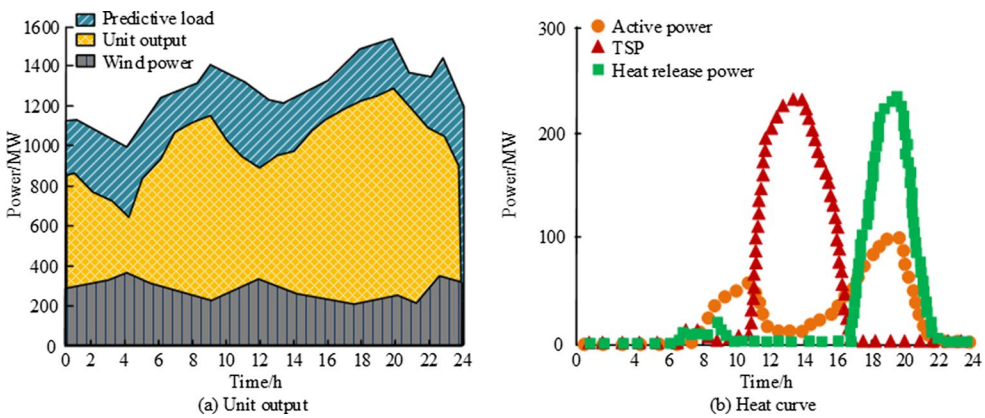

Fig. 9 Each unit output

load periods of 7:00–10:00 and 18:00–20:00, WP output is relatively low while generator output is high, which can effectively regulate the peak and valley of the power system and improve its stability and reliability. Fig. 9(b) illustrates the thermal curves against time. During the peak load period from 7:00 to 10:00, the W&S powers are relatively low, and they generate electricity through the initial capacity of the TSS, reducing conventional unit processing and thus reducing dispatch costs. The lighting power is high during the 12:00–17:00 interval, where the TSS can store more heat. STPPs reduce system scheduling costs by adjusting PG periods and thermal storage strategies. When the load is high, and WP is weak, initial thermal storage is utilized for PG. In contrast, when the light is good and energy demand is low, the system stores excess heat and reduces the amount of power it draws from the grid. Similarly, during WP peak shaving and high load, the impact is reduced by releasing heat storage.

In the end, five scenarios are set up in the experiment to compare the scheduling results, each considering the uncertainty of different combinations of WP and solar availability. After uncertainty calculation, the horizontal factor in both the robust SM and the opportunity SM is set to 0.03. Fig. 10 shows the thermal curves under two SMs. Fig. 10(a) shows the thermal curve of robust scheduling, with a TSC of 551,020 ¥. This approach is more conservative and relies heavily on conventional units, thereby increasing costs. However, it provides more stability in response to uncertainty. In contrast, Fig. 10(b) plots the thermal curve of opportunity scheduling with a TSC of 515,140 ¥. This 10% reduction in TSC demonstrates the direct cost-saving benefits of integrating MSTSPS and LBS within the IGDT-based scheduling framework. It indicates that by intelligently coordinating unit dispatch and storage operations, the system can effectively reduce scheduling costs while maintaining robust performance under uncertainty.

The experiment further sets up four scenarios to verify the effectiveness of LBS.

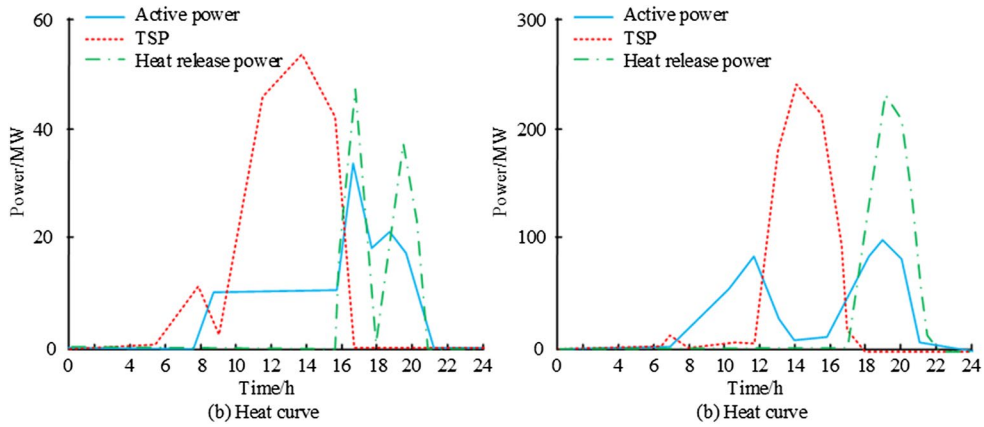

Fig. 10 Thermal curves under the two scheduling modes

Table 2 Electricity costs in the different scenarios

Scenario	**Photovoltaic** (¥)	**WP** (¥)	**TSS**(¥)	**Conventional Unit** (¥)	**Total Cost**(¥)
S1	0	0	0	42510	42510
S2	0	0	22102	16145	38247
S3	12211	8924	0	19002	40137
S4	10330	5627	652	17648	34257

Table 3 Optimization results for the different methods

Scenario	**Load Balancing**(¥)	**Coupling Correlation**(¥)	**Edge Computing**(¥)	**Requirement Response**(¥)
S1	38921	39116	40138	40317
S2	36217	37186	38546	38591
S3	35173	36179	36527	36519
S4	31247	33217	33671	34021

- **Scenario 1 (S1)** only considers conventional units. Thus, it does not include photovoltaic, WP, and TSSs.
- **Scenario 2 (S2)** considers TSS along with conventional units.
- **Scenario 3 (S3)** considers the output of photovoltaic, WP, and conventional units.
- **Scenario 4 (S4)** considers all parameters, including photovoltaic, thermal storage, WP, and conventional units.

Table 2 shows the electricity costs for different scenarios. The results show that S1 is the most expensive, with a cost of 42,510 ¥, followed by S3, with a cost of 40,137 ¥. S4, which integrates all sources, achieves the lowest cost at 34,257 ¥, highlighting the economic benefits of multi-source scheduling.

In the end, the experiment compares the coupling correlation, edge computing, and demand response scheduling methods to verify the scheduling optimization effect. Table 3 shows the optimization results of various methods. In scenarios 1 to 4, the optimization scheduling costs of LBS are 38,921 ¥, 36,217 ¥, 35,173 ¥, and 31,247 ¥, respectively, all lower than those of other methods. This indicates that LBS is highly cost-effective in scheduling optimization. The primary reason is its direct coupling with energy storage systems, which allow real-time modulation of net load profiles through peak shaving and valley filling operations. Unlike coupling correlation, which relies on statistical trend alignment between generation and demand, or edge computing approaches that prioritize localized control with limited system-wide optimization, LBS centrally orchestrates storage dispatch based on global scheduling objectives. Moreover, unlike demand response-based methods, which are often limited by user response uncertainty and require external participation, LBS enables deterministic and model-driven load shaping. This direct modulation of energy flow enables the LBS framework to respond more favorably to real-time fluctuations of renewable output and load demand, creating better cost-efficacy from the same operating conditions.

5 Conclusion

This paper presents a novel HES system for abandoned mines. The main objective of the proposed method is to improve RE integration by addressing load balancing and peak shaving challenges associated with fluctuating energy sources like wind and solar. The proposed method uses MSTSPS with LBS to mitigate variabilities in energy output. Furthermore, it uses IGDT to optimize energy distribution under uncertain conditions. The data verified with

IEEE-39 nodes showed that the proposed system reduces the PVD difference significantly, from 120 MW to 90 MW, and achieves a 10% decrease in scheduling costs. Experiments have shown that the HES system for abandoned mines can effectively balance the load demand of the power system and improve its stability and reliability through MSTSPS and LBS scheduling. Furthermore, it compares different scheduling strategies, where LBS has the lowest optimization scheduling cost and higher economic efficiency. Overall, the proposed system reduces reliance on traditional power sources and makes the grid more stable and sustainable. It is worth noting that while the proposed system demonstrates operational cost benefits, the current model does not incorporate capital investment or maintenance costs associated with storage and renewable infrastructure. Future work will address these factors through a detailed techno-economic analysis.

Furthermore, while there is the option of reusing abandoned mines with subsurface energy storage, practical implementation is subject to various technical constraints, including water intrusion risk, structural stability, and ventilation limitations. Local geotechnical and hydrogeological surveys would also be required to determine long-term storage thermal performance, strength, and viability. The cost of compliance with regulations and retrofitting needs to be added to future implementation.

Author Contributions L. Zhang led the conceptualization, methodology design, and simulation implementation. H. Ji contributed to data analysis, result interpretation, and manuscript drafting. Both authors reviewed and approved the final version of the manuscript.

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Data Availability This study did not generate or use any external datasets. All simulation results and relevant data are included within the article. No additional data are available.

Declarations

Competing Interests The authors declare that they have no competing interests.

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